Politecnico di Milano

School of Industrial and Information Engineering
Master of Science in Space Engineering
Course: Space Propulsion (A.Y. 2025–2026)

ERRATUM

Stark-100k Solid Rocket Motor

Cooling Jacket Thermal Analysis

Students

Gabriele Pisciotta
Lorenzo Veronese
Paolo Mornato
Simone Frescura
Matteo Risi Mazzola
Edouard Desmedt
Tommaso Santoro

Group: I Maggici

Abstract. The thermal analysis code submitted with the original report contained an error in the derivation of the nozzle throat area A_t. The stagnation pressure used in the computation $A_t = \dot{m}_{\text{prop}}\, c^*/p_0$ was taken from an outdated version of the ballistic properties computation code rather than from the current, validated ballistic design output. The outdated pressure value ($\approx 67\,\text{bar}$) is higher than the initial chamber pressure $P_{c,\text{in}} = 63.77\,\text{bar}$ produced by the current ballistic design, causing the throat area to be under-estimated by approximately 5 % ($A_t = 97.6\,\text{cm}^2$ instead of $102.6\,\text{cm}^2$). After correcting the input pressure and re-running the cooling-jacket sizing optimisation, new design parameters were obtained: $t_{\text{TBC}} = 72\,\mu\text{m}$ (previously 77 µm) and $\dot{m}_w = 7.6\,\text{kg/s}$ (previously 7.9 kg/s). All temperature limits and safety criteria remain satisfied. The qualitative conclusions of the original report are unchanged.

Contents

1 Source and Nature of the Error **3**
1.1 Location in the original script 3
1.2 Numerical consequence on the throat geometry 3

2 Updated Nozzle and Channel Geometry **3**

3 Updated Thermal Results **4**
3.1 Heat transfer coefficients 4
3.2 Wall heat flux 4
3.3 Temperature distribution across all wall layers 5

4 Updated Design Optimisation Parameters **5**
4.1 TBC thickness parametric sweep 5
4.2 TBC versus no-TBC comparison 6

5 Updated Safety Margins **7**
5.1 Cavitation, boiling onset, critical heat flux, and mechanical margin 7

6 Complete Summary of Updated Results **9**

7 Assessment of Conclusions **10**

1. Source and Nature of the Error

1.1 Location in the original script

The thermal analysis code computes the throat area as

$$A_t = \frac{\dot{m}_{\text{prop}}\, c^*}{p_0} \tag{1}$$

where $\dot{m}_{\text{prop}}$ is the propellant mass flow rate and c^* is the characteristic velocity from NASA CEA (Chemical Equilibrium with Applications). In the original submission, the pressure p_0 in Eq. (1) was taken from an outdated version of the ballistic properties computation code, yielding a stagnation pressure of approximately 67 bar.

The current, validated ballistic design gives an initial chamber pressure $P_{c,\text{in}} = 63.77\,\text{bar}$, which is the correct value to use in Eq. (1): A_t is a fixed geometric dimension sized at the initial operating point where the required thrust $F = 100\,\text{kN}$ is achieved.

1.2 Numerical consequence on the throat geometry

The two pressure values differ by approximately 5 %, producing the discrepancy shown in Table 1. Because channel widths scale with the local nozzle circumference (variable-width strategy, Section 3.3.2 of the original report), the smaller throat radius in the original code produced narrower channels throughout the jacket, altering the hydraulic diameter, the Gnielinski convective coefficient, and consequently the optimised design parameters.

Table 1. Geometric quantities: original (erroneous) vs. corrected values, and resulting changes in the optimised cooling design.

Quantity	Symbol	Original	Corrected
Nozzle throat — direct consequence of pressure error			
Pressure used for A_t	p_0	$\approx$67 bar	63.77 bar
Throat area	A_t	97.6 cm^2	102.6 cm^2
Throat radius	R_t	55.7 mm	57.1 mm
Throat diameter	D_t	111.5 mm	114.3 mm
Channel width at throat	$w_{\text{ch}}^{\text{th}}$	2.39 mm	2.49 mm
Channel width at inlet/outlet	$w_{\text{ch}}^{\text{in}}$	4.00 mm	4.17 mm
Optimised cooling design — re-run with corrected geometry			
TBC thickness	t_{TBC}	77 µm	72 µm
Coolant mass flow	$\dot{m}_w$	7.9 kg/s	7.6 kg/s
Channel height	h_{ch}	1.6 mm	1.5 mm
Tank pressure	p_{tank}	64 bar	52 bar

All other parameters like convergent/divergent half-angles, fin thickness, number of channels, TBC conductivity (TBC: Thermal Barrier Coating) material limits, safety factors, feed-system architecture are unchanged.

2. Updated Nozzle and Channel Geometry

The throat radius increases from 55.7 mm to 57.1 mm, uniformly widening the coolant channels along the jacket. The convergent section grows from 23.1 mm to 23.7 mm and the divergent from 86.2 mm to 88.4 mm; the total jacket length increases accordingly from 109 mm

to 112 mm. Because the channel width $w_{\mathrm{ch}}(x)$ scales proportionally with the local nozzle circumference (variable-width strategy, Section 3.3.2 of the original report), all 90 channels are uniformly 4 % wider than in the original script, with the throat width increasing from 2.39 mm to 2.49 mm and the inlet/outlet width from 4.00 mm to 4.17 mm. The corrected nozzle geometry entries replace Table 4.1 and Table 4.7 of the original report; the updated values are consolidated in Table 1 (Section 1) and Table 3 (Section 6).

3. Updated Thermal Results

3.1 Heat transfer coefficients

Figure 1 shows the axial profiles of the hot-gas-side coefficient (HTC: Heat Transfer Coefficient) h_g (Bartz) and the effective coolant-side coefficient $h_{c,\mathrm{eff}}$ (Gnielinski with fin efficiency). The corrected geometry produces channels approximately 4 % wider at all axial stations. The wider hydraulic diameter partially offsets the reduced coolant velocity, so the net change in $h_{c,\mathrm{eff}}$ is below 3 % throughout. The Bartz coefficient h_g changes by less than 2 %.

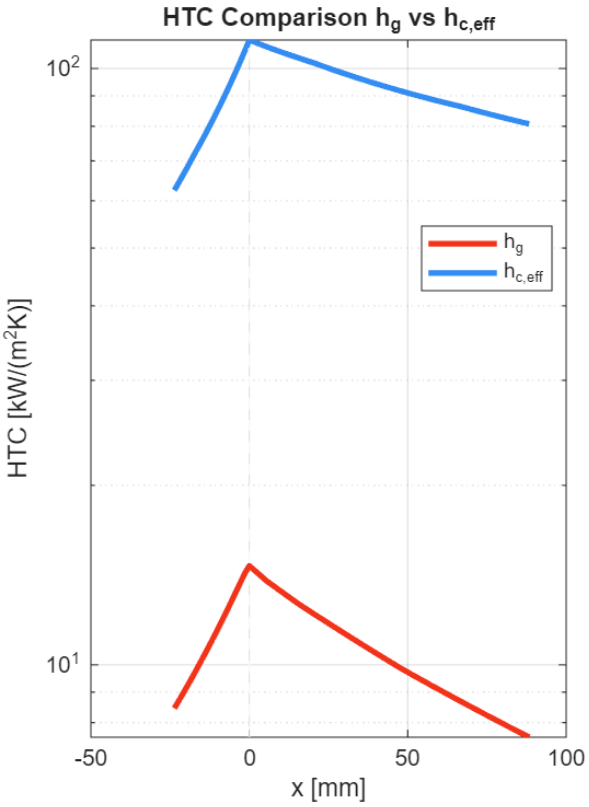


Figure 1. Corrected heat transfer coefficient profiles. h_g (Bartz) vs $h_{c,\mathrm{eff}}$ (Gnielinski, semi-log scale);

3.2 Wall heat flux

Figure 2 shows the corrected wall heat flux profile. The peak value at the throat is $q_{\max} = 19.46\,\mathrm{MW/m^2}$, compared to 19.36 MW/m^2 in the original script (difference $< 0.5\,\%$, well within the Bartz $\pm 20\,\%$ model uncertainty). The radiative contribution from CO_2/H_2O gas bands ($\varepsilon_g = 0.10$) remains below 3 % at all axial locations.

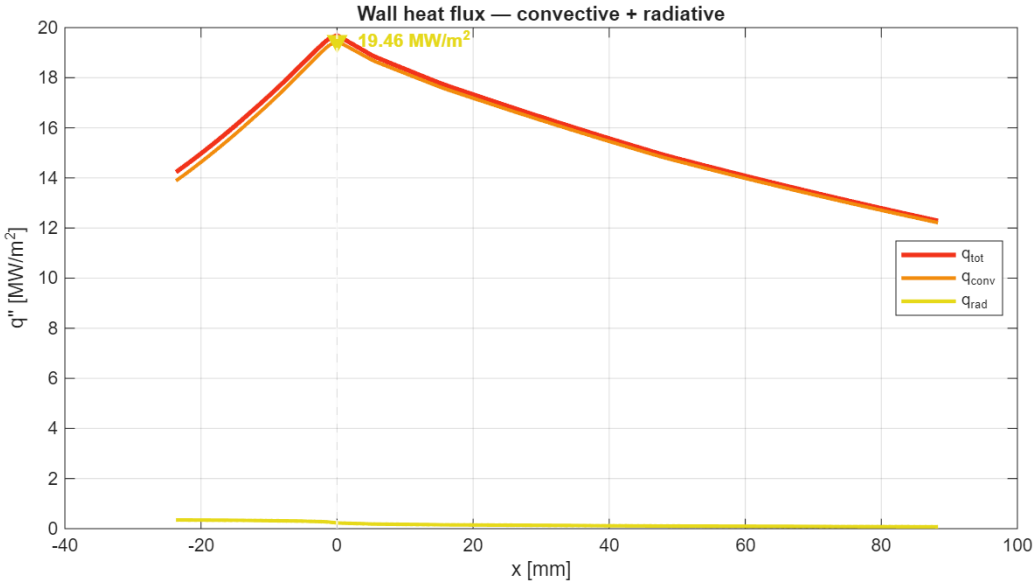


Figure 2. Corrected wall heat flux: total q_{tot}, convective q_{conv}, and radiative q_{rad} contributions. Peak: $q_{\max} = 19.46\,\text{MW}/\text{m}^2$ at the throat ($x = 0$). Replaces Fig. 4.5 of the original report.

3.3 Temperature distribution across all wall layers

Figure 3 presents the corrected axial temperature profiles across all wall layers. Updated peak values are listed in Table 3. All layers remain well below their operational limits. The slight increase in coolant temperature rise ($\Delta T_w = 26.4\,\text{K}$ vs $24.2\,\text{K}$) reflects the reduced mass flow rate ($7.6\,\text{kg/s}$ vs $7.9\,\text{kg/s}$) absorbing approximately the same total heat power.

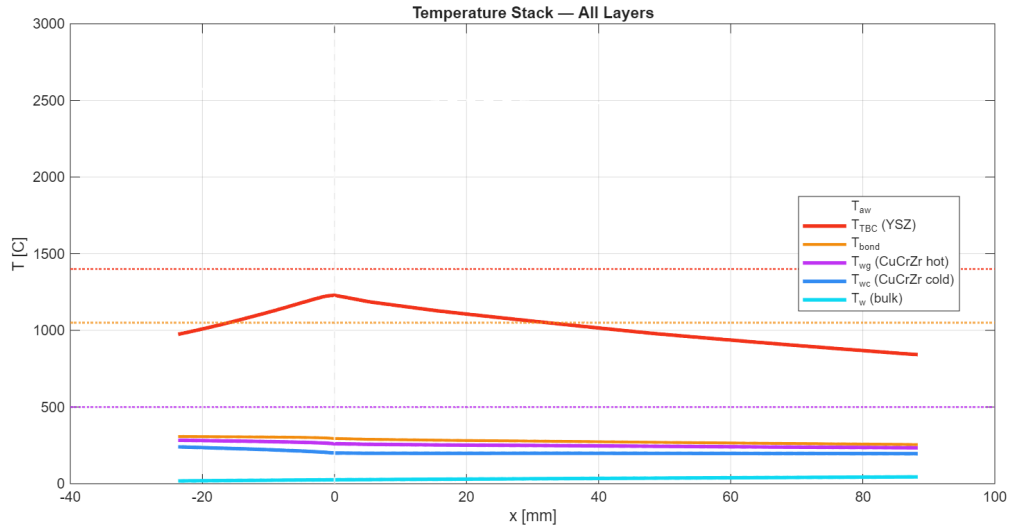


Figure 3. Corrected axial temperature stack for all wall layers. Dotted horizontal lines indicate material limits. Replaces Fig. 4.11 of the original report.

4. Updated Design Optimisation Parameters

4.1 TBC thickness parametric sweep

Figure 4 shows the corrected parametric sweep. The design point shifts from 77 µm to 72 µm, providing a 171 K margin below the YSZ (Yttria-Stabilized Zirconia) phase-stability limit ($T_{\text{TBC,max}} = 1229\,^\circ\text{C}$ vs limit $1400\,^\circ\text{C}$) and a 239 K margin for the CuCrZr liner ($T_{wg,\max} =$

261 °C vs limit 500 °C). Both margins are larger than in the original script (118 K and 220 K respectively). The qualitative shape of the sweep is unchanged.

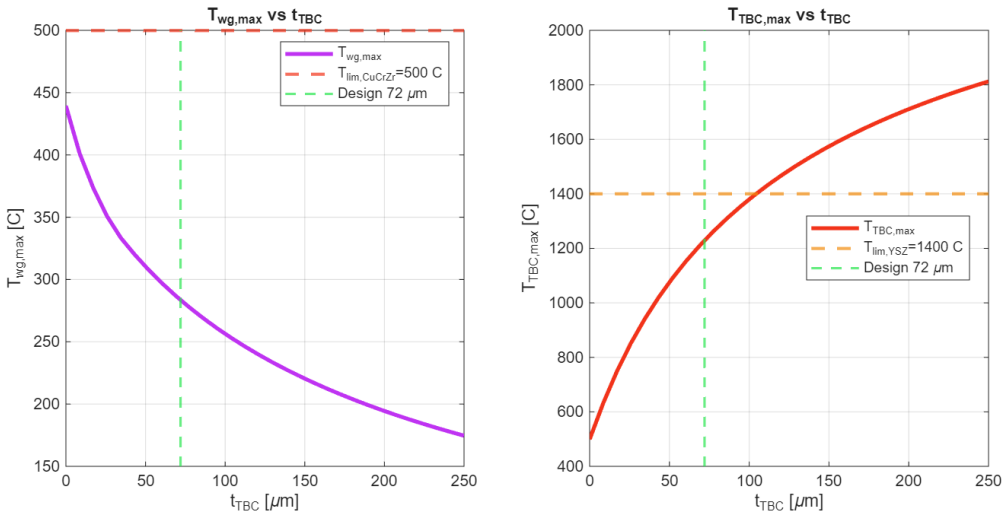


Figure 4. Corrected TBC parametric sweep. Left: $T_{wg,\max}$ vs t_{TBC}; right: $T_{\mathrm{TBC},\max}$ vs t_{TBC}. Green dashed line: corrected design point at 72 µm. Replaces Fig. 4.6 of the original report.

4.2 TBC versus no-TBC comparison

Figure 5 confirms that the YSZ TBC remains necessary with the corrected geometry. Without TBC, T_{wg} reaches $\approx 445\,^{\circ}\mathrm{C}$ at the throat (original: 456 °C), leaving only a 55 K margin. The TBC reduces the peak heat flux by 16.27 MW/m^2 (46%), compared to 17.79 MW/m^2 (48%) in the original script.

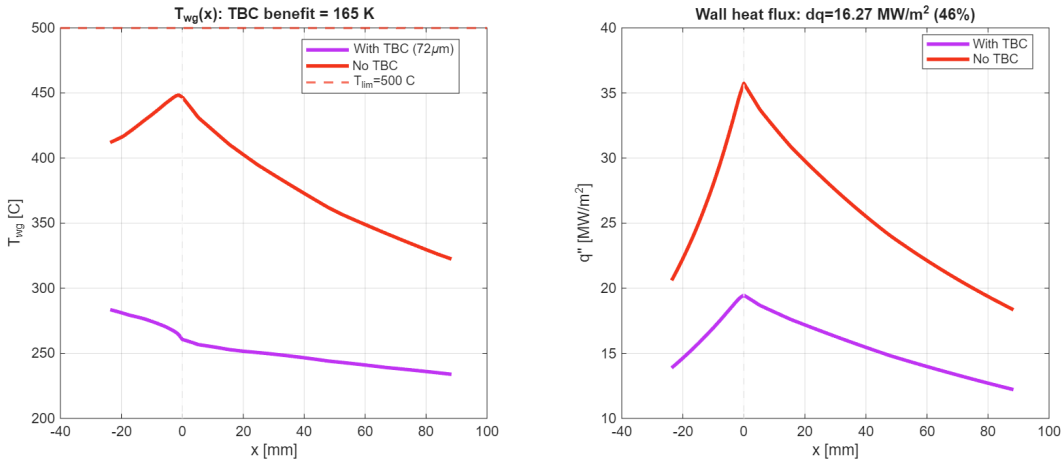


Figure 5. Corrected TBC vs no-TBC comparison: $T_{wg}(x)$ (left) and wall heat flux (right). TBC benefit: $\Delta T = 165\,\mathrm{K}$, $\Delta q = 16.27\,\mathrm{MW/m^2}$ (46%). Replaces Fig. 3.1 of the original report.

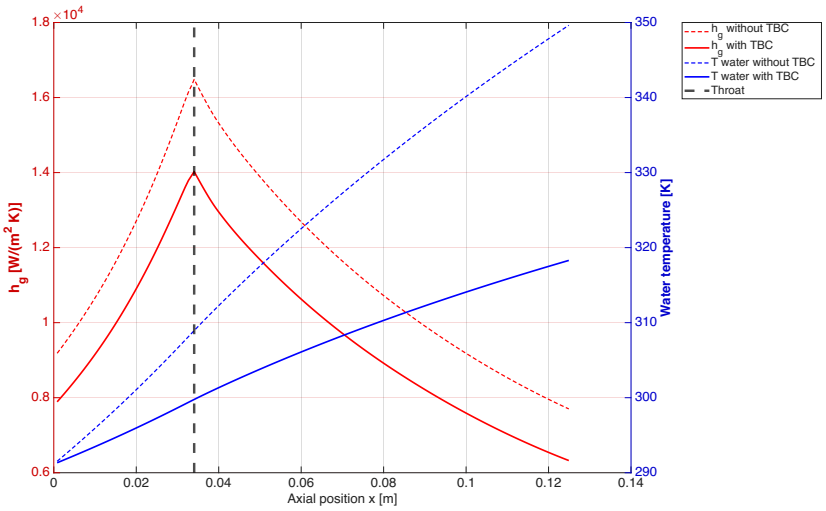


Figure 6. Corrected TBC vs no-TBC comparison of h_g and water marching temperature. Replaces Fig. 4.7 of the original report.

5. Updated Safety Margins

5.1 Cavitation, boiling onset, critical heat flux, and mechanical margin

Figure 7 shows all four safety margin profiles and Table 2 provides the quantitative comparison.

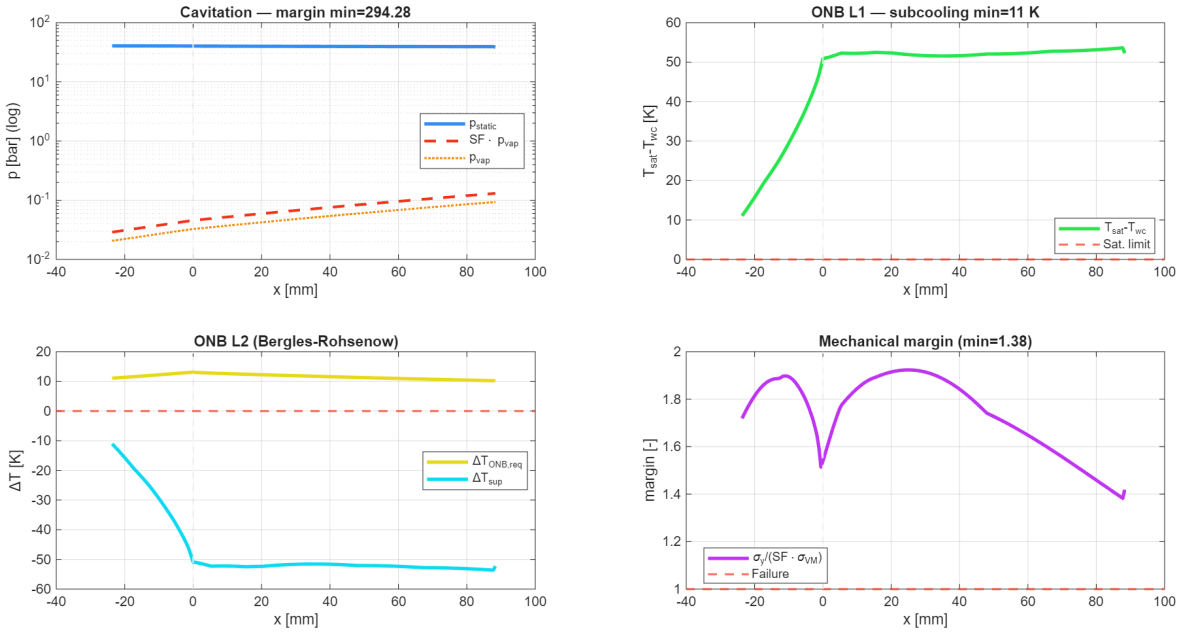


Figure 7. Corrected safety margins. Top-left: cavitation ($p_{\text{static}}/(SF_{\text{cav}}\, p_{\text{vap}})$). Top-right: ONB (Onset of Nucleate Boiling) Criterion L1 ($T_{\text{sat}} - T_{wc}$, subcooling). Bottom-left: ONB Criterion L2 — Bergles-Rohsenow threshold ($\Delta T_{\text{ONB,req}}$ vs wall superheat ΔT_{sup}). Bottom-right: Von Mises mechanical margin $\sigma_y/(\text{SF} \cdot \sigma_{\text{VM}})$. Replaces Figs. 4.8, 4.9, and 4.10 of the original report.

Table 2. Safety margin comparison. All criteria pass in both versions.

Criterion	Requirement	Original	Corrected	Status
Cavitation: $p_{\text{static}}/(SF \cdot p_{\text{vap}})$ min	> 1	425.7	294.3	**PASS**
ONB L1: $T_{\text{sat}} - T_{wc}$ (min) [K]	> 0	$+30.2$	$+11.0$	**PASS**
ONB L2: B-R ratio max	< 1	0	0	**PASS**
CHF: $q_{\text{CHF,FC}}/q_{\text{flux,ch}}$ min	> 1.5	2.2	1.9	**PASS**
Mechanical (Von Mises): $\sigma_y/(\text{SF} \cdot \sigma_{\text{VM}})$ min	> 1	1.04	1.38	**PASS**

Cavitation. The margin decreases from 425.7 to 294.3 due to the lower regulated tank pressure (52 bar vs 64 bar), which reduces p_{static} throughout the jacket. Both values are orders of magnitude above the threshold; the change has no engineering significance.

ONB L1. The minimum subcooling margin decreases from $+30.2\,\text{K}$ to $+11.0\,\text{K}$ at the convergent inlet ($x = -23\,\text{mm}$). The reduction follows directly from the lower p_{static}: at 52 bar, the saturation temperature ($T_{\text{sat}} \approx 264\,^\circ\text{C}$) is approximately 12 °C lower than at 64 bar ($T_{\text{sat}} \approx 276\,^\circ\text{C}$). The criterion is still satisfied everywhere: $T_{wc} < T_{\text{sat}}$, confirming single-phase forced convection.

ONB L2. The Bergles-Rohsenow ratio is zero in both versions because $T_{wc} < T_{\text{sat}}$ throughout the jacket (wall subcooled everywhere). The L2 criterion applies only when the wall is superheated ($T_{wc} > T_{\text{sat}}$), a condition that never occurs here. Setting $q_{\text{ONB,BR}} = \infty$ for all subcooled nodes is physically correct; a ratio of zero does not imply zero heat flux.

Critical heat flux (CHF). The conservative pool-boiling (Zuber) model gives $SF_{\text{CHF,pool}} = 1.5$ (equal to the stated requirement of 1.5). The forced-convection model (Celata enhancement factor), which is more representative of the actual operating conditions, gives $SF_{\text{CHF,FC}} > 1.5$ in both versions, confirming that the CHF criterion is met.

Mechanical margin. The minimum Von Mises margin improves from 1.04 to 1.38 at the divergent outlet, where the channel is widest. The improvement arises because the lower tank pressure reduces the differential pressure $|p_{\text{coolant}} - p_{\text{gas}}|$ across the liner, decreasing the membrane stress σ_{mem}.

6. Complete Summary of Updated Results

Table 3. Complete results: original (erroneous) vs. corrected. Bold values indicate quantities that changed.

Quantity	Symbol	Original	Corrected	Limit / Req.
Nozzle geometry				
Throat area [cm^2]	A_t	97.6	**102.6**	—
Throat radius [mm]	R_t	55.7	**57.1**	—
Jacket length [mm]	L_{jkt}	109.3	**112.1**	—
Optimised cooling design				
TBC thickness [μm]	t_{TBC}	77	**72**	—
Coolant flow [kg/s]	$\dot{m}_w$	7.9	**7.6**	—
Channel height [mm]	h_{ch}	1.6	**1.5**	—
Tank pressure [bar]	p_{tank}	64	**52**	—
Thermal jacket inlet pressure [bar]	P_{in}	51.94	**51.99**	—
Wall heat flux				
Peak heat flux [MW/m^2]	q_{max}	19.36	**19.46**	—
Radiative fraction [%]	—	1.2	$\approx$1.2	—
TBC heat-flux reduction [MW/m^2]	Δq	17.79	**16.27**	—
TBC heat-flux reduction [%]	—	47.9	**46.0**	—
Temperatures				
$T_{\text{TBC,max}}$ [°C]	—	1282	**1229**	< 1400
$T_{wg,\text{max}}$ [°C]	—	280	**261**	< 500
$T_{\text{bond,max}}$ [°C]	—	304	**295**	< 1050
T_{TBC} margin [K]	—	118	**171**	—
T_{wg} margin [K]	—	220	**239**	—
Coolant outlet temp [°C]	$T_{w,\text{out}}$	42.2	**44.4**	—
Coolant ΔT [K]	—	24.2	**26.4**	—
Safety margins				
Cavitation margin (min)	—	425.7	**294.3**	> 1
ONB L1 subcooling (min) [K]	—	$+30.2$	$+11.0$	> 0
ONB L2: B-R ratio (max)	—	0	0	< 1
CHF: SF_{FC} (min)	—	2.2	1.9	> 1.5
Mechanical margin (min)	—	1.04	**1.38**	> 1
Feed system (N_2 blow-down, adiabatic design model)				
N_2 volume — adiabatic [L]	V_{N2}	123.6	**97.1**	—
N_2 mass — adiabatic [kg]	m_{N2}	42.6	**33.5**	—
N_2 volume — isothermal (ref) [L]	—	68.6	**51.8**	—

The reduction in N_2 pressurant volume from 123.6 L to 97.1 L (adiabatic model) follows directly from the lower tank pressure: at 52 bar, the gas needs to expand less to reach the regulator lock-up point, requiring a smaller vessel.

7. Assessment of Conclusions

The geometric correction produces quantitative changes to all parameters listed in Table 3. From an engineering standpoint the following assessments hold:

1. **All safety criteria pass** with comfortable margins in the corrected design. No design revision beyond the updated optimisation parameters is required.
2. **The TBC is mandatory** regardless of the geometric correction: without TBC, $T_{wg,\max} \approx 445\,^{\circ}\mathrm{C}$ leaves only a 55 K margin against the CuCrZr limit.
3. **Single-phase forced convection** is confirmed throughout the jacket. ONB L1 is the tightest criterion ($+11\,\mathrm{K}$ at the inlet); ONB L2 is trivially satisfied because the wall is subcooled everywhere.
4. **Material selection** (CuCrZr liner, YSZ TBC, NiCoCrAlY bond coat) and **cooling architecture** (single-pass water, N_2 pressurised system) are unaffected.
5. **Sections 3.1, 3.2, and 3.4** of the original report (propellant chemistry, ballistic design, external support architecture) are not affected by this correction.